EgoBlind-RA: Towards Safer Egocentric Assistive AI for Blind Users via Risk-Adaptive Routing

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Abstract

Blind and low-vision (BLV) users depend on visual assistance systems to navigate an unpredictable physical world, yet existing multimodal systems apply a uniform inference policy irrespective of query urgency. We present **EgoBlind-RA**, a risk-adaptive framework that classifies egocentric query-video pairs as *urgent* or *non-urgent* via a lightweight CLIP-based classifer and routes each query to a response calibrated to its safety stakes. Our classifier achieves 0.905 ROC-AUC on the held-out tes set. We compare two LoRA-fine-tuned architectures of Kimi-VL-A3B-Instruct on the EgoBlind benchmark: a bifurcated design with separate urgent and non-urgent adapters, and a unified prompt-tag-conditioned design. Urgency-conditioned SFT against carefully selected reference targets substantially improves over a uniform-policy baseline in both. Surprisingly, Direct Preference Optimization regresses SFT in the unified setting whilst proving productive in the bifurcated setting, where the best result mixes an SFT urgent adapter with a DPO non-urgent adapter. This suggests the value of preference-pair-based alignment depends sharply on architectural regime.

1 Introduction

Vision-language models have advanced rapidly, yet their development has largely been shaped by users with full sensory access. For the ∼253M people worldwide living with visual impairment [1], AI systems that interpret egocentric video and answer natural-language questions about the immediate environment offer a transformative opportunity to support independence and personal safety. Recent benchmarks including EgoBlind [2] and visual question answering datasets such as VizWiz [3] have begun to formalize this setting, exposing challenges specific to blind and low-vision (BLV) users: off-centre regions of interest, ambiguous spatial context, motion blur, and information needs ranging from casual to life-critical. Performance here matters far beyond academic interest, because the cost of a misleading answer is asymmetric and, in the safety-critical case, severe.

Despite this, state-of-the-art multimodal systems apply a uniform inference policy across all queries. A user asking *"Is there a step in front of me?"* faces an immediate physical hazard demanding a fast, conservative response; a user asking *"What brand of cereal is this?"* does not. Models such as LLaVA [4] and InstructBLIP [5] treat both identically. The gap cuts in both directions: excessive caution degrades usability for routine queries, whilst insufficient urgency on safety-critical ones can cause physical harm. Crucially, query urgency is not determined by text alone — *"Is there anything in my way?"* carries different stakes at the edge of a subway platform than at home — so an effective response policy must condition on the visual scene as well as the question. To our knowledge, no prior system jointly classifies visual urgency and routes queries to risk-stratified policies in this setting.

In this paper, we propose **EgoBlind-RA**, a risk-adaptive egocentric visual assistance framework for BLV users. EgoBlind-RA is a two-stage pipeline: a lightweight CLIP-based classifier predicts

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whether each query–video pair is *urgent* or *non-urgent*, and the predicted urgency conditions a fine-tuned vision-language model. We investigate two architectures: a bifurcated design with separate urgent and non-urgent Low-Rank Adaptation (LoRA) adapters (**Approach 1**), and a unified single-adapter design conditioned by an urgency prompt-tag (**Approach 2**). Both are optimized under a regime-specific composite loss balancing accuracy, assistive utility, and generation latency. We make four contributions. First, a frozen-backbone CLIP classifier learns urgency from joint video-text representations, achieving 0.905 ROC-AUC on test. Second, urgency-conditioned SFT against carefully selected reference targets — *shortest*-correct for urgent queries, *longest*-correct for non-urgent — substantially reduces composite loss over the baseline in both variants. Third, in Approach 2, Direct Preference Optimization (DPO) [6] on top of SFT *regresses* the result, whether initialized from SFT or from the base model; diagnostic analysis traces this to a uniform verbosity drift across question types. Fourth, in Approach 1, DPO is a productive component: an urgent-only DPO model substantially outperforms the baseline, and the best overall result mixes an SFT urgent adapter with a DPO non-urgent adapter — suggesting the value of preference-pair-based alignment depends sharply on the architectural regime in which it is applied.

2 Related Work

Visual assistance for BLV users. VizWiz [3] introduced an early assitive VQA system in which BLV users photographed their surroundings and asked questions that were subsequently answered by crowd workers. The work highlighted challenges characteristic of BLV-generaed images, including poor framing, motion blur, and queries spanning both low- and high-stakes istuations. VisWiz [7] later formalized this setting as a large-scale VQA benchmark, demonstrating BLV-generated images are substantially more difficult for existing models than standard VQA inputs. More recently, EgoBlind [2] extended the problem to egocentric video; we adopt EgoBlind as our evaluation dataset. However, none of these benchmarks evaluates whether models adapt their behavior to the urgency or safety-criticality of user queries.

Egocentric video understanding. Ego4D [8] introduced a large corpus of first-person video supporting tasks such as episodic memory, action anticipation, and future forecasting. Embodied Question Answering [9] extended question answering to interactive embodied settings, where an agent must navigate through an environment to obtain the information needed to answer a query. However, both settings assume sighted users and treat all queries uniformly, without accounting for differences in urgency or safety criticality.

Vision-language models and efficient adaptation. Recent large vision-language models (VLMs), including LLaVA [4], InstructBLIP [5], Qwen-VL [10], and Kimi-VL [11], achieve strong out-of-the-box performance on visual question answering tasks. We use Kimi-VL as our base model. Our urgency classifier is built on representations from CLIP [12], while LoRA [13] enables efficient fine-tuning of multi-billion-parameter models on a single GPU.

Alignment and preference optimization. Reinforcement Learning from Human Feedback (RLHF) [14] aligns language models to human preferences using a reward model trained on pairwise comparisons. DPO [6] simplifies this process by optimizing directly from preference pairs, and is now widely adopted. Variants include Kahneman-Tversky Optimization (KTO) [15], which uses scalar utility labels instead of pairwise comparisons, and Reinforcement Learning from AI Feedback (RLAIF) [16], which replaces human labels with AI feedback. We apply DPO with urgency-conditioned preference pairs and find that it can underperform standard SFT, a failure mode not previously reported for short, vision-grounded VQA.

Risk-aware inference. Several lines of work adapt model behavior to input risk: models can abstain when uncertain [17], estimate uncertainty using semantic entropy [18], or allocate more compute to difficult inputs [19]. However, these approaches are text-only and do not incorporate a learned visual urgency signal.

How our work differs. To our knowledge, we are the first to classify urgency in egocentric video queries and route them to response policies conditioned on risk level. Prior BLV work provides datasets and evaluation protocols, but does not study risk-aware inference. Prior alignment work

generally treats DPO as a reliable improvement over SFT; in contrast, we show that its effect depends strongly on architecture, degrading performance in our unified-adapter setup whilst improving the best-performing bifurcated-adapter model.

3 Problem Statement

Let $\mathbf{x} = (v, q)$ denote a video-query pair, where $v \in \mathcal{V}$ is an egocentric video clip and $q \in \mathcal{Q}$ is a natural language query from a BLV user. Each pair has a binary urgency label $u \in \{0, 1\}$ and a reference answer set $\mathcal{Y}^* = \{y_1^*, \ldots, y_K^*\}$. EgoBlind-RA solves two coupled subproblems:

(1) learn an **urgency classifier** $f_\theta : \mathcal{V} \times \mathcal{Q} \to \{0, 1\}$;

(2) learn a **conditional response generator** $g_\phi : \mathcal{V} \times \mathcal{Q} \times \{0, 1\} \to \mathcal{A}$, whose objective depends on the predicted urgency $\hat{u} = f_\theta(\mathbf{x})$. We instantiate g in two ways: Approach 1 trains separate generators g_{ϕ_u}, g_{ϕ_n}, and selects between them by $\hat{u}$; Approach 2 trains a single generator g_ϕ that takes $\hat{u}$ as an input prompt-tag.

The objective for g is regime-specific: a composite loss balancing accuracy, assistive utility, and generation latency for urgent queries, and an accuracy-and-utility-only loss for non-urgent ones. We define these objectives in Section 4.3.

4 Proposed Method

EgoBlind-RA operates as a two-stage pipeline, illustrated in Figure 1: (i) an urgency classifier predicts the risk level of each query, and (ii) a conditional response generator produces an answer whose objective is matched to the predicted urgency regime. We investigate two instantiations of the second stage.

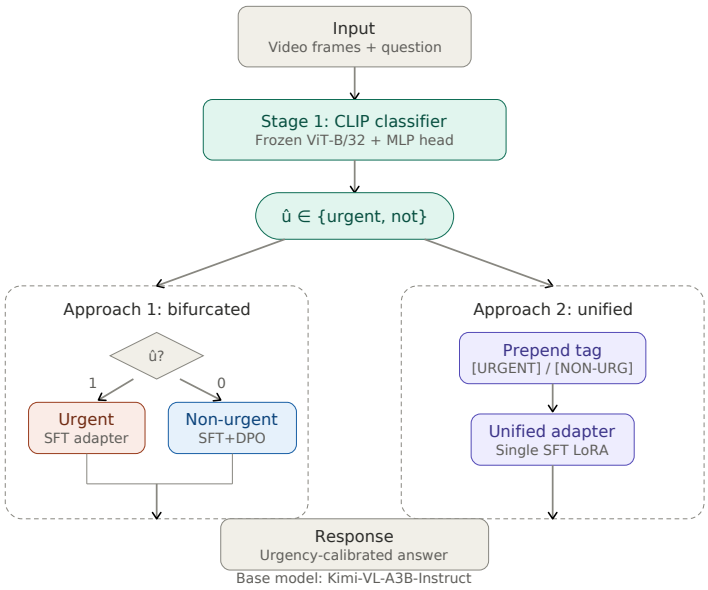

Figure 1: EgoBlind-RA pipeline. A CLIP-based classifier predicts query urgency from the joint video–text representation; the predicted urgency then routes to one of two architectural variants. Approach 1 selects between separate urgent and non-urgent LoRA adapters. Approach 2 prepends an urgency tag to a single unified adapter. Both are scored under the same regime-specific composite loss.

4.1 Stage 1: Urgency Classification

Given a video–query pair $\mathbf{x} = (v, q)$, we train a lightweight classifier f_θ to predict the binary urgency label $\hat{u} \in 0, 1$. The model is built on a frozen CLIP ViT-B/32 backbone, where frame and text embeddings are fused and fed into a two-layer MLP head trained with binary cross-entropy loss.

We keep the classifier lightweight so it can act as a fast preprocessing step before the response generator. Freezing the backbone avoids end-to-end fine-tuning of CLIP while preserving its strong multimodal representations. At inference time, we use a default decision threshold of 0.5; threshold tuning to reduce false negatives is discussed as future work in Section 7. Additional architectural and training details appear in Section 5, with full results in Section 6.1.

4.2 Stage 2: Urgency-Conditioned Response Generation

We compare two architectural instantiations of the response generator (Figure 1). Both approaches use the same composite loss from Section 3; they differ only in how urgency is incorporated into the generator.

Approach 1: Modular hard-routing (bifurcated adapters). We train two separate LoRA adapters on a shared base model $\mathcal{M}$. The *urgent adapter* ϕ_u is fine-tuned on the subset $\mathcal{D}_u = \{(\mathbf{x}_i, y_i) : u_i = 1\}$ using the full loss $\mathcal{L}_{\text{urgent}}$, which includes the latency penalty $\mathcal{L}_{\text{lat}}$ to encourage concise, actionable responses. Training targets are the *shortest* correct reference answer for each example, biasing the model toward brevity. The *non-urgent adapter* ϕ_n is fine-tuned on $\mathcal{D}_n = \{(\mathbf{x}_i, y_i) : u_i = 0\}$ using only $\mathcal{L}_{\text{acc}} + \mathcal{L}_{\text{util}}$, with the *longest* reference answer as target to encourage thorough, detailed responses. At inference time, the classifier prediction $\hat{u}$ determines which adapter is loaded:

$$y = \begin{cases} \mathcal{M}_{\phi_u}(\mathbf{x}) & \text{if } \hat{u} = 1, \\ \mathcal{M}_{\phi_n}(\mathbf{x}) & \text{if } \hat{u} = 0. \end{cases} \tag{1}$$

This hard-routing design allows each adapter to specialize its generation behavior without compromising the other regime.

Approach 2: Prompt-conditioned unified adapter. A single LoRA adapter ϕ_{uni} is trained on the full dataset $\mathcal{D}u \cup \mathcal{D}n$. Urgency is provided as a control tag pre-pended to the system prompt. The loss switches accordingly: $\mathcal{L}$urgent for `[URGENT]` examples and $\mathcal{L}$non-urgent for `[NON-URGENT]` examples (Section 4.3). This setup evaluates whether prompt conditioning alone enables a single model to learn risk-adaptive behavior without architectural changes.

4.3 Composite Loss

The composite loss is regime-specific. For an urgent query ($\hat{u} = 1$), the response y is optimized under

$$\mathcal{L}_{\text{urgent}} = \alpha\, \mathcal{L}_{\text{acc}} + \beta\, \mathcal{L}_{\text{util}} + \gamma\, \mathcal{L}_{\text{lat}}, \qquad \alpha + \beta + \gamma = 1, \quad \alpha, \beta, \gamma > 0. \tag{2}$$

For a non-urgent query ($\hat{u} = 0$), the latency term is dropped and the remaining weights are renormalized:

$$\mathcal{L}_{\text{non-urgent}} = \frac{\alpha}{\alpha + \beta}\, \mathcal{L}_{\text{acc}} + \frac{\beta}{\alpha + \beta}\, \mathcal{L}_{\text{util}}. \tag{3}$$

We define each term below.

Accuracy. $\mathcal{L}_{\text{acc}}$ penalizes semantic mismatch between y and the closest reference in $\mathcal{Y}^*$:

$$\mathcal{L}_{\text{acc}}(y, \mathcal{Y}^*) = 1 - \max_{k \in [K]} S(y, y_k^*), \tag{4}$$

where $S(\cdot, \cdot) \in [0, 1]$ is token-overlap F1. The max over references rewards matching any valid answer, accommodating EgoBlind's up-to-four-answer reference sets.

Assistive utility. EgoBlind provides a $[0, 5]$ utility score measuring whether the answer is actionable for a BLV user beyond lexical correctness. We normalize and treat its complement as a loss:

$$\mathcal{L}_{\text{util}}(y, \mathcal{Y}^*) = 1 - \frac{\text{Score}(y, \mathcal{Y}^*)}{5}. \tag{5}$$

A verbose paraphrase and a terse direct answer can score similarly under S but differ sharply in utility, motivating a separate term.

Latency. For urgent queries, a correct answer arriving too late is no longer useful. We use response token count $|y|$ as a differentiable proxy for generation latency. The penalty is piecewise: zero below $\tau_{\min}$ (to avoid rewarding one-word answers), polynomial growth between $\tau_{\min}$ and $\tau_{\max}$ controlled by exponent p, and a steep linear penalty beyond $\tau_{\max}$ with slope κ (closed form in Appendix C). $\mathcal{L}_{\text{lat}}$ is dropped entirely for non-urgent queries, where verbosity is acceptable. This asymmetric treatment is what makes the loss *risk-adaptive* rather than merely accuracy-adaptive.

4.4 Optimization and Inference

Optimization. Both approaches are trained using SFT with parameter-efficient adapters [13], guided by an urgency-conditioned composite loss.

In Approach 1, separate urgent and non-urgent adapters are trained on disjoint datasets $\mathcal{D}_u$ and $\mathcal{D}_n$, each optimized under its respective loss. The urgent adapter is supervised using the shortest correct reference response, while the non-urgent adapter is supervised using the longest. In Approach 2, a single adapter is trained on the union $\mathcal{D}_u \cup \mathcal{D}_n$, with the loss conditionally applied per example via a prepended urgency tag. In both cases, this supervision strategy biases the model toward concise, actionable outputs for safety-critical queries and more detailed responses for routine ones.

For the urgent adapter in Approach 1, and for the shared adapter on $\mathcal{D}_u$ in Approach 2, we further apply DPO [6] to align outputs with the urgent composite reward. Preference pairs are constructed by sampling k candidate responses per prompt, scoring each, and selecting the highest-scoring response as the "chosen" and the lowest-scoring as the "rejected." Additional details on scoring, filtering, and the choice of k are provided in Section 5.

Inference. Both approaches begin inference identically: given a video–query pair $\mathbf{x} = (v, q)$, a CLIP-based classifier predicts urgency $\hat{u} = f_\theta(\mathbf{x}) \in \{0, 1\}$ using a 0.5 threshold on the MLP output.

In Approach 1, $\hat{u}$ selects which adapter is activated (Eq. 2), and the response is generated greedily from the corresponding adapter atop the shared base model. In Approach 2, $\hat{u}$ instead determines whether a `[URGENT]` or `[NON-URGENT]` tag is prepended to the system prompt, while a single shared adapter generates the response.

Thus, the two approaches represent a trade-off between parameter modularity and prompting-based control: Approach 2 reduces storage requirements by using a single adapter but relies on prompt conditioning for behavioral separation, whereas Approach 1 enforces separation structurally through distinct adapters.

5 Experimental Methodology

Dataset. We use the EgoBlind dataset [2], which consists of egocentric video clips paired with natural-language questions from BLV users, each accompanied by ≤ 4 reference answers per question. EgoBlind organizes each question into 7 categories: information reading, safety warnings, navigation, social communication, tool use, communication & interaction, and other resources. Since the dataset does not include urgency annotations, we augment it by labeling each question as either urgent or non-urgent from the perspective of a BLV user. To do so, we prompt GPT-5.2 (`gpt-5.2-2025-12-11`, `reasoning_effort=xhigh`) with five sampled frames per video clip together with the corresponding question (full prompt in Appendix B). We validate this annotation procedure by manually reviewing a random subset of 300 questions, finding over 90% agreement with the model-generated labels.

Data splits. The training set contains 2,746 questions across 922 videos (1,474 urgent and 1,272 non-urgent), while the test set consists of 1,283 questions (668 urgent and 615 non-urgent). To train the CLIP-based urgency classifier, we further split the training data into stratified 80/20 train and validation sets to preserve the class distribution.

Uniform-policy baseline. We evaluate a uniform-policy baseline using `kimi-vl-a3b-instruct` (Moonshot API). For each test question, we provide frames extracted from the corresponding video together with the question text, and use a single urgency-agnostic system prompt instructing the model to respond as a blind person answering a first-person query (full prompt in Appendix A). The same prompt is applied to all inputs, regardless of urgency, yielding a strict uniform-policy

baseline against which urgency-aware fine-tuning is compared. We use a temperature of 0.2 and set `max_tokens` to 256.

CLIP urgency classifier. The classifier is built on a frozen `CLIP ViT-B/32` backbone (OpenAI weights). For each query timestamp t_q, we sample $K = 4$ frames uniformly from a ± 2-second window around t_q, mean-pool their CLIP image embeddings, and concatenate the result with the CLIP text embedding of the question. A two-layer MLP head takes this fused representation and predicts urgency. The CLIP backbone is frozen; only the MLP is trained, with AdamW (lr $= 10^{-4}$), binary cross-entropy loss, and 5 epochs on a single NVIDIA L40S GPU.

Fine-tuning. We fine-tune Kimi-VL-A3B-Instruct [11] via LoRA [13] on MIT's Engaging cluster (NVIDIA L40S GPUs) for both architectural variants.

Approach 1 (bifurcated adapters).
Two separate LoRA adapters (rank 8, $\alpha = 16$, targeting `q_proj` and `v_proj`) are trained on urgency-stratified subsets using the `kimi_vl_nothink` template. The urgent adapter trains on $\mathcal{D}_u$ ($n = 1{,}474$) with AdamW (lr $= 1 \times 10^{-4}$), batch size 1, gradient accumulation 8, and 1 epoch in bf16; each example includes one video frame and the shortest reference answer as target. The non-urgent adapter trains on $\mathcal{D}_n$ ($n = 1{,}272$) with lr $= 2 \times 10^{-5}$ (the higher rate caused NaN divergence due to longer targets) and the longest reference answer as target. DPO pairs are generated by sampling $k = 4$ responses, scoring by token-overlap F1, and selecting the best/worst as chosen/rejected ($\beta = 0.1$, lr $= 5 \times 10^{-6}$, 3 epochs). For the non-urgent adapter, DPO is applied via stacked adapters: the SFT adapter is merged into the base before the DPO adapter is applied as a delta.

Approach 2 (unified adapter). A single LoRA adapter (rank 8, $\alpha = 16$, dropout 0.05, targeting `q_proj` and `v_proj`) is trained under 4-bit quantization with AdamW (cosine schedule, lr 1×10^{-4}, warmup ratio 0.1, weight decay 0.01), per-device batch size 1, gradient accumulation over 8 steps, and 1 epoch in bf16. The dataset is vision-conditioned (`cutoff_len` 2048) with urgency tags prepended to each example; for urgent questions the SFT target is the *shortest* correct reference answer, and for non-urgent questions the *longest*. We also experiment with three DPO variants on top of the SFT checkpoint: (i) composite-loss-scored preference pairs ($k = 8$ candidates, three-filter selection, 975 retained pairs); (ii) token-overlap-F1-scored pairs from the SFT checkpoint; and (iii) token-overlap-F1-scored pairs from the base model. All DPO runs use $\beta = 0.1$, sigmoid loss, learning rate 5×10^{-6}, and 3 epochs.

Evaluation metrics. For the urgency classifier we report accuracy, precision, recall, F1, and ROC-AUC on both validation and test partitions. For the response generator we report the composite loss $\mathcal{L}$ along with mean output length and similarity (token-overlap F1 against the closest reference) on the EgoBlind test set. We score under the canonical grid-search-optimal hyperparameters $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$, $\tau_{\min} = 8$, $\tau_{\max} = 40$, $p = 3$, $\kappa = 0.3$, selected by minimizing overall test loss across 3,888 configurations (486 satisfying $\alpha + \beta + \gamma = 1$).

6 Results and Discussion

6.1 CLIP Urgency Classifier

Results. Table 1 reports classifier performance on the held-out validation and test sets. Despite a frozen backbone, the model learns a strong urgency signal, achieving ROC-AUC scores of 0.938 (validation) and 0.905 (test), indicating robust discriminative performance across splits. The main limitation is recall on the urgent class (0.695 on test), reflecting a bias toward under-predicting urgency, which is particularly costly in safety-critical settings.

Discussion. The strong ROC-AUC across both splits indicates that the frozen CLIP backbone captures a meaningful urgency signal from the joint video–text representation without end-to-end fine-tuning. The performance gap between validation (86.3%) and test (79.8%) is expected given that the test set contains unseen videos. The main limitation is recall on urgent cases: the model misses approximately 30% of urgent queries on the test set, misclassifying them as non-urgent, which is the most costly error in a safety-critical setting. Improving recall via threshold tuning, asymmetric loss weighting, or partial backbone unfreezing is left for future work (Section 7).

Table 1: CLIP urgency classifier performance.

Metric	Validation	Test
Accuracy	0.863	0.798
Precision	0.930	0.879
Recall	0.807	0.695
F1 Score	0.864	0.777
ROC-AUC	0.938	0.905

6.2 Baseline Evaluation

Table 5 reports overall performance: a composite loss of 0.607, with a mean output length of 84.1 words and a similarity score of 0.350. While the baseline is not inaccurate per se (the similarity is non-trivial), it is substantially over-verbose, producing responses exceeding 80 words for questions whose reference answers are typically only a few words long. As a result, the latency component of the composite loss dominates, indicating that prompt-only urgency conditioning is insufficient to induce concise responses for safety-critical queries.

Per-category analysis. Table 6 breaks down the best-configuration composite loss by question type. Safety warnings have the highest loss (1.118), followed by tool use (0.948) and navigation (0.787). Information reading and social communication, which are predominantly non-urgent, exhibit the lowest losses. This pattern aligns with EgoBlind [2], which finds that models perform worst on categories where errors are most consequential, further motivating urgency-aware approaches.

Composite loss grid search. We evaluate the baseline across 3,888 hyperparameter configurations of the composite loss $(\alpha, \beta, \gamma, \tau_{\min}, \tau_{\max}, p, \kappa)$, of which 486 satisfy the normalization constraint $\alpha + \beta + \gamma = 1$. Among constrained settings, the best test loss is achieved with $\alpha = 0.5, \beta = 0.3, \gamma = 0.2, \tau_{\min} = 8, \tau_{\max} = 40, p = 3, \kappa = 0.3$.

The top 5 configurations share $\alpha = 0.5, \beta = 0.3, \gamma = 0.2$ and $\tau_{\max} = 40$, differing only in $\tau_{\min}$ and p, suggesting relative insensitivity to latency-regime boundaries but sensitivity to the accuracy–latency trade-off weights. Notably, the optimal $\gamma = 0.2$ assigns non-trivial weight to latency, indicating that brevity is important and supporting our design choice to decouple latency penalties for non-urgent queries. We use this configuration as the canonical scoring throughout the paper.

6.3 SFT Results

6.3.1 Approach 1

Training the bifurcated adapters required resolving two issues beyond the template fix shared with Approach 2. First, text-only training (no video frames) produced degenerate outputs across all configurations: the best text-only adapter generated "Yes." or "No." for 71% of urgent queries, pattern-matching common short references without visual grounding. Including a single video frame per example resolved this. Second, aggressive LoRA configurations (rank 16, all projection targets, $\geq$2 epochs) consistently overfitted; adopting the lighter configuration from Approach 2 stabilized training. Full ablation details appear in Appendix E.

With vision-conditioned SFT, the urgent adapter achieves 0.505 similarity on urgent test examples and the non-urgent adapter achieves 0.268 on non-urgent examples. Under oracle routing (ground-truth urgency labels), the combined system achieves 0.399 overall similarity, roughly $4\times$ the uniform-policy baseline (0.103), while reducing mean output length from 84.1 to 4.0 words.

6.3.2 Approach 2

Two implementation issues initially impeded vision-conditioned fine-tuning. First, the LLaMA-Factory [20] training pipeline applied a default chat template incompatible with Kimi-VL's native format, producing mis-tokenized inputs during training and degenerate outputs at inference. We resolve this by registering a custom `kimi_vl_nothink` template that matches the model's expected dialogue structure. Second, the released `modeling_kimi_vl.py` is incompatible with `transformers` ≥ 4.50

across several API mismatches; we patch these in both the downloaded model and the runtime module cache via an idempotent setup script.[2] On applying these fixes, optimization becomes stable and the training loss decreases monotonically.

Table 4 reports SFT against the uniform-policy baseline. SFT reduces composite loss from 0.607 to 0.375 (a 38.1% relative improvement), drops mean output length from 84.1 to 2.2 words, and raises reference similarity from 0.350 to 0.589. Urgency-conditioned SFT therefore enforces concise outputs without sacrificing accuracy — in fact, brevity and similarity improve together.

6.4 DPO Results

6.4.1 Approach 1

We evaluate four DPO configurations: initialized from SFT or from the base model, for each urgency regime. Results are summarized in Table 2.

Table 2: Approach 1 initial training method comparison (token-overlap F1 similarity on test set, oracle routing).

Method	Urgent	Non-urg.	Overall
Baseline (uniform policy)	0.124	0.080	0.103
SFT only	0.505	0.268	0.391
DPO only (from base)	0.277	0.270	0.273
SFT followed by DPO	0.459	0.298	0.381
Best mix (SFT urgent, SFT+DPO non-urgent)	0.507	0.298	0.407

For the urgent adapter, DPO on SFT degrades performance (0.505 $\rightarrow$ 0.459), consistent with the regression observed in Approach 2. DPO from base improves over the untrained model (0.124 $\rightarrow$ 0.277) but remains far below SFT. For the non-urgent adapter, DPO on SFT yields a modest gain (0.268 $\rightarrow$ 0.298). The best overall result *mixes* the SFT urgent adapter with the SFT + DPO non-urgent adapter (0.407 overall).

Cross-architecture comparison. DPO degrades the urgent adapter in both approaches for the same reason: urgent reference answers are typically 1-3 words long, so sampled preference pairs contain minimal variation, and DPO has little meaningful signal to learn from. For non-urgent queries, reference answers are substantially longer and more varied, giving DPO enough contrast between chosen and rejected responses to produce a genuine improvement. This explains why the bifurcated design can benefit from DPO (its non-urgent adapter trains exclusively on long-answer examples) while the unified design cannot (its DPO pairs mix both regimes, and the short-answer urgent examples dilute the signal).

End-to-End Pipeline Evaluation We evaluate the full Approach 1 pipeline by replacing oracle routing with the CLIP classifier.

Table 3: End-to-end Approach 1 pipeline under different routing strategies (test set, $n = 1{,}283$).

Routing strategy	Loss	Similarity	Mean words
Baseline (uniform policy)	3.247	0.103	84.1
Zero-shot (prompted Kimi-VL)	0.481	0.346	5.1
CLIP classifier	**0.436**	**0.393**	4.5
Oracle (ground-truth labels)	0.421	0.399	4.0

A zero-shot baseline (prompting base Kimi-VL to classify urgency) achieves only 0.45% recall on urgent queries, effectively collapsing the bifurcated design. The trained CLIP classifier achieves 78.1% accuracy and 65.9% recall (F1 = 0.759), raising pipeline similarity from 0.346 (zero-shot) to

[2] Specifics in the project repository.

0.393, closing 88% of the gap to oracle routing (0.399). The remaining gap is attributable to 228 false negatives, urgent queries misrouted to the non-urgent adapter. The classifier, not just the response generators, is also a primary pipeline bottleneck.

6.4.2 Approach 2

We apply DPO [6] on top of the SFT checkpoint to align the urgent-regime adapter with the urgency-aware composite reward. We train three variants to assess whether DPO improves over SFT, and whether any gains are robust to the choice of pair scoring and initialization.

Composite-loss-scored pairs. For each urgent training example, we sample $k = 8$ responses from the SFT model (temperature 0.7) and score them using $\mathcal{L}_{\text{urgent}}$. The lowest-loss response is treated as the "chosen" and the highest-loss as the "rejected." We apply three filtering criteria to remove low-quality pairs: identical responses, loss_gap < 0.2, and yes/no polarity mismatches. This yields 975 preference pairs. DPO is trained with $\beta = 0.1$, a sigmoid loss, learning rate 5×10^{-6}, and 3 epochs.

Token-overlap F1 pairs. We also consider a simpler scheme where $k = 4$ sampled responses are scored using token-overlap F1 against the longest reference. The highest-scoring response is used as the "chosen" and the lowest as the "rejected." We only filter identical-candidate pairs. We evaluate this setup from both the SFT checkpoint and the base model to isolate the effect of initialization on DPO performance.

Table 4: Final composite-loss comparison across all methods on the test set ($n = 1{,}283$). Across the three DPO variants tested, all regress the SFT result, regardless of pair-scoring method or initialization. Mean words and similarity are reported alongside loss to characterize the regression. Percentages are computed from unrounded values.

Method	Loss	Δ vs SFT	Mean words	Similarity
Baseline (few-shot prompted Kimi-VL)	0.607	$+61.6\%$	84.1	0.350
SFT (vision LoRA)	**0.375**	—	2.2	0.589
SFT $\rightarrow$ DPO (composite-loss pairs)	0.433	$+15.5\%$	7.3	0.522
SFT $\rightarrow$ DPO (token-overlap F1 pairs)	0.526	$+40.1\%$	18.2	0.424
Baseline $\rightarrow$ DPO (token-overlap F1 pairs)	0.543	$+44.7\%$	25.2	0.407

Results. Every DPO variant regresses SFT (Table 4). Composite-loss DPO from SFT increases loss by 15.5% (0.375 $\rightarrow$ 0.433); F1 DPO from SFT lands at +40.1% (0.526); F1 DPO from the base model is worst at +44.7% (0.543). The regression is robust to pair-scoring method and to initialization.

Discussion. Two diagnostic signals explain the regression. First, verbosity drift is uniform across question types: mean output length jumps from 2.2 (SFT) to 7.3 words, with near-identical drift across yes/no ($+4.8$), what ($+5.2$), where ($+5.9$), and how ($+6.8$) questions. DPO learned a global "be wordy" preference, not a question-conditional length policy. Second, the reward components fight each other: training pairs pointed chosen toward higher similarity *and* shorter length, yet the DPO model produces both longer outputs and lower similarity than SFT, degrading on both axes despite preferences pointing the opposite way.

For short-form egocentric VQA where references provide strong supervision, SFT against the shortest correct reference is a stronger objective than DPO. SFT enforces brevity per example, while DPO only ranks one short response above one long one. The failure generalizes across pair-scoring schemes, with F1-scored DPO regressing further than composite-loss DPO, so it is not specific to our reward formulation.

7 Conclusion and Future Directions

We present **EgoBlind-RA**, a risk-adaptive framework for egocentric BLV visual assistance. The system combines a CLIP-based urgency classifier (test ROC-AUC = 0.95) with two urgency-conditioned

LoRA-fine-tuned variants of Kimi-VL-A3B-Instruct: a bifurcated design with separate urgent and non-urgent adapters, and a unified design conditioned via prompt tags.

In the bifurcated-adapter setting (Approach 1), oracle routing achieves a roughly $4\times$ improvement in similarity over the baseline, and the trained CLIP classifier closes 88% of the gap between naïve zero-shot routing and oracle routing, yielding a deployable pipeline within 1.5% of the oracle ceiling. Unlike Approach 2, DPO is partially productive: it degrades the urgent adapter but improves the non-urgent one, and the best overall result mixes SFT and DPO adapters. This asymmetry suggests that preference-pair alignment requires regime-homogeneous training data, a condition satisfied by bifurcated adapters but not by unified training.

In the unified-adapter setting (Approach 2), urgency-conditioned SFT against carefully selected reference targets reduces composite loss by 38.1% over a uniform-policy baseline, with mean output length decreasing from 84.1 to 2.2 words and similarity increasing from 0.350 to 0.589. In contrast, applying DPO on top of SFT consistently degrades performance across pair-scoring strategies and initialization schemes. Our analysis attributes this regression to a uniform verbosity shift across question types and to misalignment between similarity and latency components under pairwise preference optimization.

These results suggest that, in short-form egocentric VQA settings with strong reference supervision, carefully constructed SFT objectives provide a more effective training signal than preference-pair-based alignment methods.

Limitations. Our findings come with several limitations. We use token-overlap F1 as a proxy for semantic similarity; while scalable, metrics such as BERTScore or LLM-based judges would provide more reliable estimates but are computationally prohibitive at our scale. The CLIP-based urgency classifier is trained with standard binary cross-entropy and a fixed 0.5 decision threshold; however, in a safety-critical assistive setting, false negatives are more costly than false positives, and we do not explicitly incorporate this asymmetry during training or calibration. Finally, our evaluation does not include human-in-the-loop studies with blind and low-vision (BLV) users, which is necessary to validate real-world utility.

For Approach 1, the primary limitation is classifier dependence: the CLIP classifier misses $\sim$30% of urgent queries, routing them to the non-urgent adapter. While the system degrades gracefully (a correct but verbose response), the added latency could matter for physical safety. The bifurcated design also doubles adapter storage relative to Approach 2, though both adapters are lightweight.

Future directions. For Approach 1, threshold tuning to favor recall over precision would directly improve urgent-query handling, since misrouting urgent queries is more costly than the reverse. Multi-frame inference (matching the classifier's four-frame setup) may improve response grounding. Finally, Group Relative Policy Optimization (GRPO) [**?**], which maintains per-example supervision, may offer a more principled alternative to DPO for the urgent regime.

For Approach 2, three extensions are particularly relevant. First, improving urgency classification is critical, as the CLIP classifier currently misses 30% of urgent queries on the test set. This may be addressed via threshold tuning, asymmetric loss weighting to penalize false negatives more heavily, and limited fine-tuning of upper CLIP layers. Second, on the alignment side, the observed regression of DPO suggests the need for methods that retain per-example supervision rather than relying on pairwise preferences. Candidates include reward-weighted SFT, which directly reweights training examples by composite score, and distillation from a stronger teacher model. Finally, we have not evaluated the full pipeline end-to-end. In particular, classifier errors can misroute queries into the wrong response regime, potentially compounding downstream failures. Future work should quantify this cascading effect in terms of safety impact, not only component-level performance.

8 Code and Reproducibility

The implementation of **EgoBlind-RA** is publicly available on GitHub at github.com/juliavekim/EgoBlind-RA. Individual course repositories, containing course assignments and the EgoBlind-RA final project, are also available: Julia's and Xander's.

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A Baseline Prompt

The uniform-policy baseline uses a single system prompt for every test question, regardless of the GPT-5.2-assigned urgency label. The prompt is adapted from the EgoBlind benchmark's evaluation setup [2].

```
I want you to act as a blind person. I will provide you with
a question raised by a blind person from their first-person
perspective in the current scene. Your task is to answer the blind
person's question. The answer needs to be based on the content of
the picture and the objective characteristics that the blind person
cannot see. If the question cannot be answered, you can say I don't
know. Do not include Chinese characters in your response. The
question is: {question}
```

B Urgency Annotation Prompt

We annotate the EgoBlind dataset with urgency labels using GPT-5.2 (`gpt-5.2-2025-12-11`, `reasoning_effort=xhigh`). For each question, the model receives five evenly-spaced video frames alongside the question text, and is prompted to return either `urgent` or `not_urgent`. The complete system prompt is reproduced below.

```
You are a classifier for the EgoBlind dataset, which contains
egocentric video from blind or visually impaired people navigating
the real world. You must classify each question as 'urgent' or
'not_urgent'.

A question is URGENT if reducing response latency would significantly
improve safety, prevent physical harm, or prevent irreversible
loss or missed opportunity -- meaning speed is more important than
perfect accuracy in this situation.

Classify as URGENT when:
  • The person is about to take an immediate physical action
    (stepping, crossing, touching, boarding, pouring, cutting,
    exiting a vehicle, etc.)
  • The answer will immediately influence the person's next physical
    movement or decision
  • The environment is dynamic or changing (moving vehicles, closing
    doors, approaching obstacles, active traffic)
  • There is imminent risk of injury (traffic, falling, tripping,
    burns, sharp objects)
  • There is risk of losing something or missing a time-sensitive
    opportunity (vehicle leaving, item left behind, boarding ending)
Classify as NOT_URGENT when:
  • Accuracy is more important than speed
  • The question is descriptive or exploratory
  • It involves reading text, identifying medications or products,
    or verifying information where correctness matters more than
    immediacy
  • It concerns planning or orientation without imminent movement or
    time pressure
  • Delay would not meaningfully change the outcome
Use the video context to determine whether the situation involves
immediate action, dynamic change, or time-sensitive consequences.

Respond with ONLY one word: urgent or not_urgent.
```

The user-side prompt accompanying each set of frames is:

```
These are frames from an egocentric video recorded by a blind
person (start time: {start_time}s). They are asking the following
question:

Question: {question}

Is this person in an immediate action-risk situation where delay
increases danger, or is this a time-sensitive question where a slow
response makes the answer useless? Respond with ONLY: urgent or
not_urgent.
```

C Loss Components

The closed form of the latency penalty referenced in Section 4.3 is

$$\mathcal{L}_{\text{lat}}(y) = \begin{cases} 0 & \text{if } |y| \le \tau_{\min} \\ \left(\dfrac{|y| - \tau_{\min}}{\tau_{\max} - \tau_{\min}}\right)^p & \text{if } \tau_{\min} < |y| \le \tau_{\max} \\ 1 + \kappa\,(|y| - \tau_{\max}) & \text{if } |y| > \tau_{\max} \end{cases}, \tag{6}$$

where $p \ge 1$ controls curvature in the transition zone and $\kappa > 0$ is the overflow slope. The grid-searched values used throughout the paper are $\tau_{\min} = 8$, $\tau_{\max} = 40$, $p = 3$, $\kappa = 0.3$.

D Tables

Table 5: Uniform-policy baseline (`moonshot-v1-32k-vision-preview`) performance on the test set under the constrained-optimal grid-search configuration.

Metric	Value	N
Composite loss	0.607	1,283
Mean output length	84.1 words	—
Similarity (token F1)	0.350	—

Table 6: Per-category composite loss under the constrained-optimal grid-search configuration ($\alpha = 0.5, \beta = 0.3, \gamma = 0.2, \tau_{\min} = 8, \tau_{\max} = 40, p = 3, \kappa = 0.3$; baseline, test set).

Question Type	Loss	Similarity	N
Safety warnings	1.118	0.118	946
Tool use	0.948	0.139	211
Navigation	0.787	0.117	363
Other resources	0.501	0.112	222
Information reading	0.489	0.089	1,873
Social communication	0.459	0.065	37
Communication & interaction	0.450	0.065	81

E Approach 1 Training Ablations

Table 7 summarizes the progression of Approach 1 training configurations. The first three runs used text-only inputs; all produced degenerate outputs despite low training loss. SFT v1 used the default `kimi_vl` template, which injects thinking tokens (◁`think`▷/◁`/think`▷) that corrupted generation. SFT v2–v3 fixed the template but retained aggressive LoRA settings (rank 16, all 7 projection targets), causing catastrophic overfitting. SFT v4 adopted the lighter configuration from Approach 2 (rank 8, `q_proj+v_proj`, 1 epoch) and produced coherent text-only outputs, but 71% of urgent predictions collapsed to "Yes." or "No."; pattern-matching without visual grounding. Vision-conditioned SFT

resolved all issues. The non-urgent adapter additionally required a $5\times$ learning-rate reduction ($1 \times 10^{-4} \rightarrow 2 \times 10^{-5}$); at the higher rate, gradient norms diverged to NaN within the first epoch.

Table 7: Approach 1 SFT ablation summary. All runs use Kimi-VL-A3B-Instruct.

Run	**Template**	**LoRA**	**Vision**	**Epochs**	**Outcome**
v1	`kimi_vl`	rank 16, all 7	No	5	Thinking-token garbage
v2	`nothink`	rank 16, all 7	No	2	Repetition loops
v3	`nothink`	rank 16, all 7	No	2	Still degenerate
v4	`nothink`	rank 8, q+v	No	1	Yes/No collapse (71%)
Final	`nothink`	rank 8, q+v	**Yes**	1	**Coherent (0.505 sim)**